Quantitative

Group 6

2026-08-18

Results

Summary Stats

- Find rating distributions in pg: 5–7.
- Find list of highest to lowest rated questions in pg: 7–8.

Inter-Rater Reliability

Inter-rater reliability was assessed using Krippendorff's α (ordinal) across six raters for GPT-4, five raters for GPT-5, and eight evaluation criteria. Overall agreement was below the commonly accepted threshold of 0.667 [@krippendorff] for both models (GPT-4 $\alpha = 0.372$, GPT-5 $\alpha = 0.117$; pg: 11). Agreement varied considerably across criteria: Ambiguity and Narrativity showed near-zero or negative α values for both models, indicating systematic disagreement, while Repetition showed the highest agreement for GPT-4 ($\alpha = 0.543$), approaching but not reaching the acceptable threshold. GPT-5 agreement remained low across all criteria (pg: 11–12). A sensitivity analysis excluding each rater individually showed that no single rater was responsible for the low agreement — the most impactful exclusion (R3) improved GPT-4 α from 0.372 to 0.535 and GPT-5 α from 0.117 to 0.216 (pg: 12–15). This suggests the disagreement is structural, likely reflecting the inherent subjectivity of certain criteria rather than individual rater error. Given this, all five raters were retained for subsequent analyses.

Paired samples t-test

The Irrelevance criterion was excluded from inferential analyses due to a skewed distribution (ceiling effect), as inspection of its histogram revealed ratings concentrated at the higher end of the scale, violating the normality assumption.

A paired samples t-test was conducted to compare ratings between GPT-4 ($M = 3.31$, $SD = 0.949$) and GPT-5 ($M = 3.35$, $SD = 0.955$) across seven criteria. No significant overall difference was found ($t(565) =$ -1.04,$p = .299$, mean diff = -0.049, 95% CI [-0.143, 0.044]).

Per-criterion analyses were conducted with Bonferroni correction applied($\alpha = .05/7 = .007$). Two criteria showed significant differences after correction: GPT-4 was rated significantly higher on Narrativity ($t(74) = 5.22$, $p < .001$, $d = 0.603$), while GPT-5 was rated significantly higher on Factuality ($t(74) =$ -3.20, $p = .014$, $d =$ -0.370). Both represent medium effect sizes. Recall and Reduction showed uncorrected significant differences that did not survive Bonferroni correction (both $p > .30$ after correction).

A limitation of this study is the small number of raters ($n = 5$). However, given the Central Limit Theorem, the aggregated ratings are expected to approximate a normal distribution, supporting the validity of parametric testing.

Data analysis R code

Preprocessing

Libraries

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggpubr)
library(irr)
```

```
## Loading required package: lpSolve
```

```
library(effectsize)
```

```
## Warning: package 'effectsize' was built under R version 4.6.1
```

```
library(jsonlite)
```

```
##
## Attaching package: 'jsonlite'
##
## The following object is masked from 'package:purrr':
##
##     flatten
```

Get individual CSVs

```
files <- list.files(pattern = "\\.csv$")
```

Append into one CSV and convert to long format

```r
complete <- lapply(files, function(f) {
  read.csv(f) %>%
    mutate(Rater = tools::file_path_sans_ext(basename(f)))
}) %>%
  bind_rows() %>%
  relocate(Rater) %>%
  pivot_longer(
    cols = -c(Rater, Question, Model),
    names_to = "Criteria",
    values_to = "Rating"
    )
# make sure ratings are numeric values
complete$Rating <- as.numeric(complete$Rating)
```

View complete data

```r
view(complete)
head(complete)
```

```
## # A tibble: 6 x 5
##   Rater Question Model Criteria       Rating
##   <chr> <chr>    <chr> <chr>           <dbl>
## 1 A     Q1       GPT4  Recall              2
## 2 A     Q1       GPT4  Factuality          4
## 3 A     Q1       GPT4  Narrativity         3
## 4 A     Q1       GPT4  Reduction           3
## 5 A     Q1       GPT4  Generalization      3
## 6 A     Q1       GPT4  Ambiguity           4
```

Summary

```r
summary(complete)
```

```
##        Rater           Question           Model           Criteria
##  Length   :3000   Length   :3000   Length   :3000   Length   :3000
##  N.unique :   6   N.unique :  24   N.unique :   3   N.unique :   8
##  N.blank  :   0   N.blank  :   0   N.blank  :   0   N.blank  :   0
##  Min.nchar:   1   Min.nchar:   2   Min.nchar:   3   Min.nchar:   6
##  Max.nchar:   3   Max.nchar:   3   Max.nchar:   4   Max.nchar:  14
##
##
##      Rating
##  Min.   :0.000
##  1st Qu.:3.000
##  Median :4.000
##  Mean   :3.934
##  3rd Qu.:5.000
##  Max.   :5.000
##  NAs    :817
```

```r
# grouped by model and criteria
complete %>% group_by(Model, Criteria) %>%
  summarise(across(Rating, list(mean=mean, sd=sd, min=min, max=max),
                   na.rm = TRUE))
```

```
## Warning: There was 1 warning in `summarise()`.
## i In argument: `across(...)`.
## i In group 1: `Model = "GPT4"`, `Criteria = "Ambiguity"`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
##   # Previously
##   across(a:b, mean, na.rm = TRUE)
##
##   # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Model and Criteria.
## i Output is grouped by Model.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Model, Criteria))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## # A tibble: 24 x 6
## # Groups:   Model [3]
##    Model Criteria       Rating_mean Rating_sd Rating_min Rating_max
##    <chr> <chr>                <dbl>     <dbl>      <dbl>      <dbl>
##  1 GPT4  Ambiguity             3.14     0.821          0          5
##  2 GPT4  Factuality            2.84     0.934          0          5
##  3 GPT4  Generalization        3.05     0.825          0          5
##  4 GPT4  Irrelevance           4.30     1.07           0          5
##  5 GPT4  Narrativity           3.45     0.836          0          5
##  6 GPT4  Recall                3.34     0.971          0          5
##  7 GPT4  Reduction             2.99     0.707          0          4
##  8 GPT4  Repetetion            3.76     1.15           0          5
##  9 GPT5  Ambiguity             3.31     0.930          1          5
## 10 GPT5  Factuality            3.2      0.885          1          5
## # i 14 more rows
```

Rating distribution per criteria

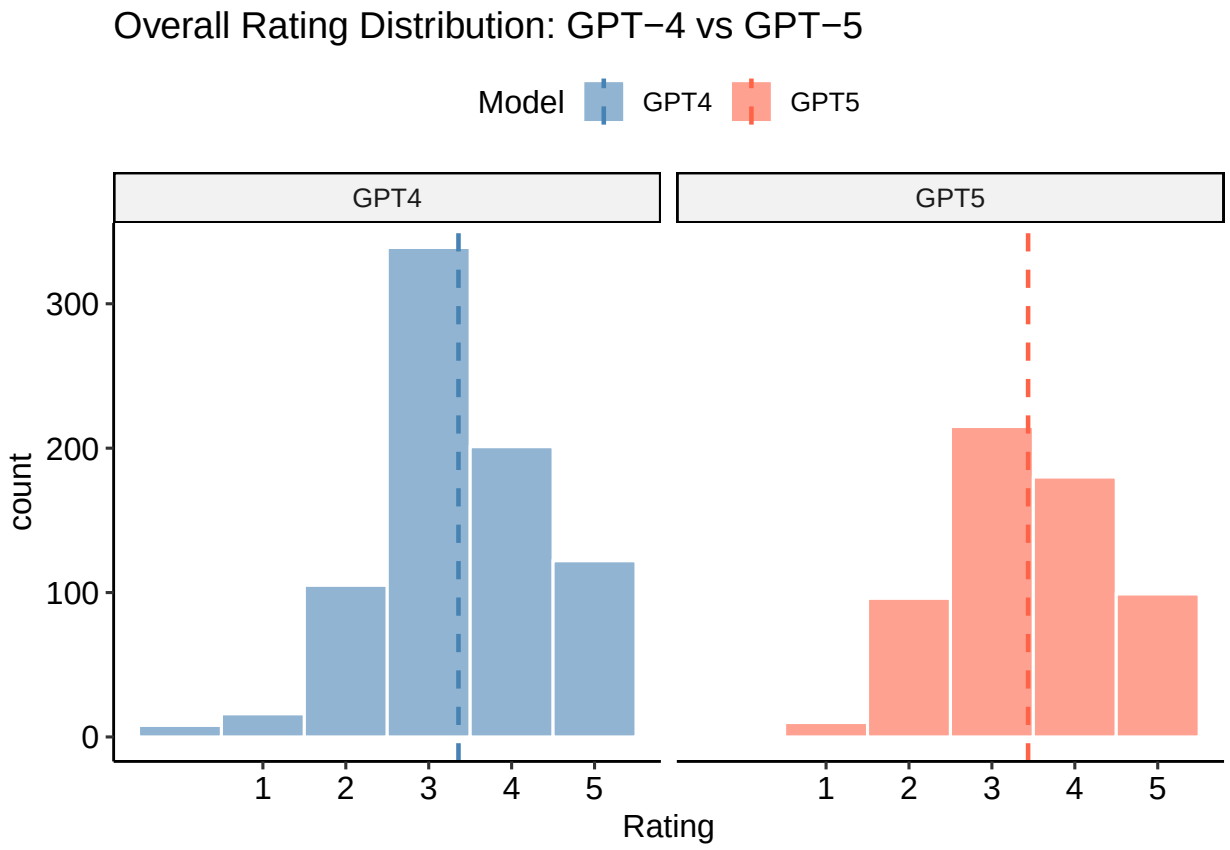
- check if data is normally distributed

```r
complete.filtered <- complete %>%
  filter(Model %in% c("GPT4", "GPT5")) %>%
  filter(!is.na(Rating))

# Overall overlay histogram (all criteria combined)

ggplot(complete.filtered, aes(x = Rating, fill = Model)) +
```

```
  geom_histogram(
    binwidth = 1,
    boundary = -0.5,
    color = "white",
    alpha = 0.6
  ) +
  facet_wrap(~Model) +
  scale_x_continuous(breaks = c(1, 2, 3, 4, 5)) +
  scale_fill_manual(values = c("GPT4" = "steelblue", "GPT5" = "tomato")) +
  geom_vline(
    data = complete.filtered %>%
      group_by(Model) %>%
      summarise(mean = mean(Rating, na.rm = TRUE)),
    aes(xintercept = mean, color = Model),
    linewidth = 0.8, linetype = "dashed"
  ) +
  scale_color_manual(values = c("GPT4" = "steelblue", "GPT5" = "tomato")) +
  theme_pubr() +
  ggtitle("Overall Rating Distribution: GPT-4 vs GPT-5")
```




```
plot <- function(data, title) {
    ggplot(data, aes(x = Rating)) +

    geom_histogram(
      binwidth = 1,
```

```
      boundary = -0.5,
      fill = "steelblue",
      color = "white"
    ) +

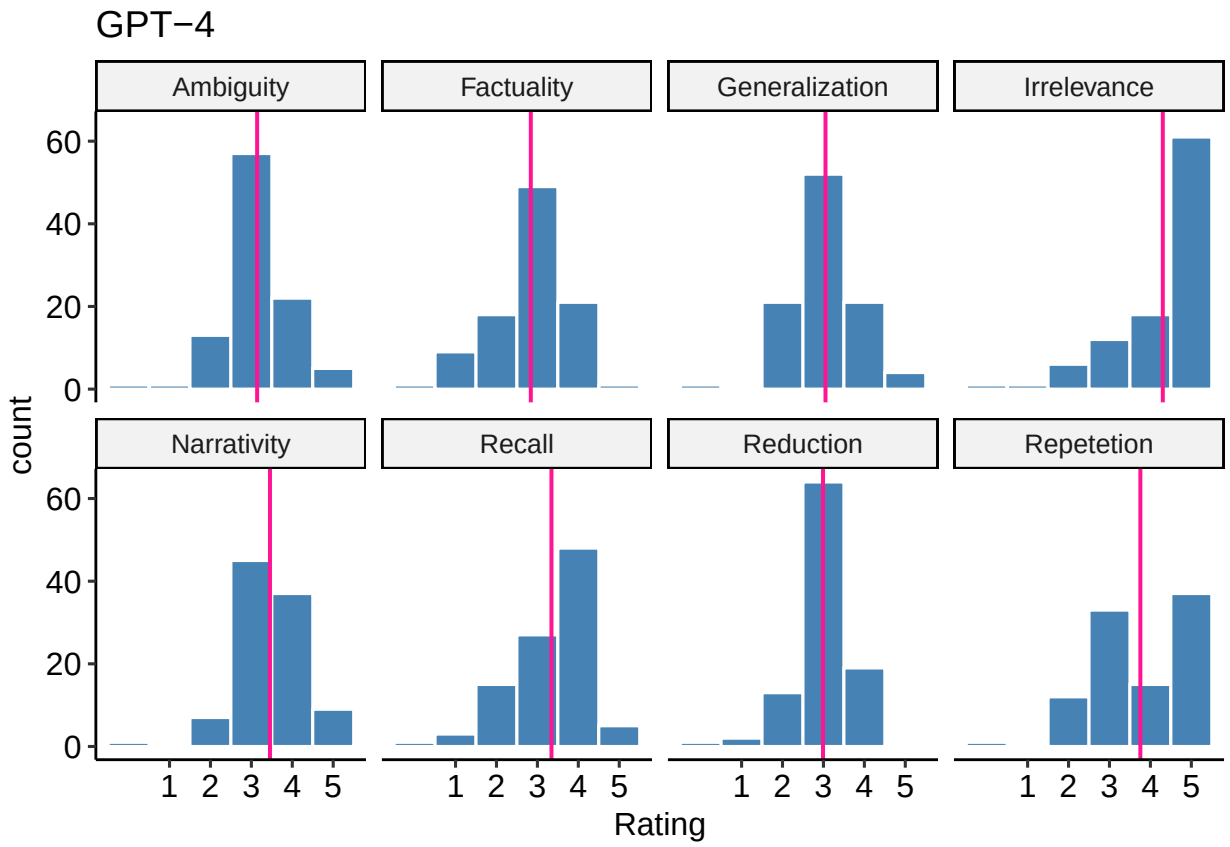
    scale_x_continuous(
      breaks = c(1, 2, 3, 4, 5)
      ) +

    geom_vline(
      data = data %>%
        group_by(Criteria) %>%
        summarise(mean = mean(Rating, na.rm = TRUE)),
      aes(xintercept = mean),
      colour = "deeppink",
      linewidth = 0.7
    ) +

    facet_wrap(~Criteria, ncol=4) +
    theme_pubr() +
    ggtitle(title)
}

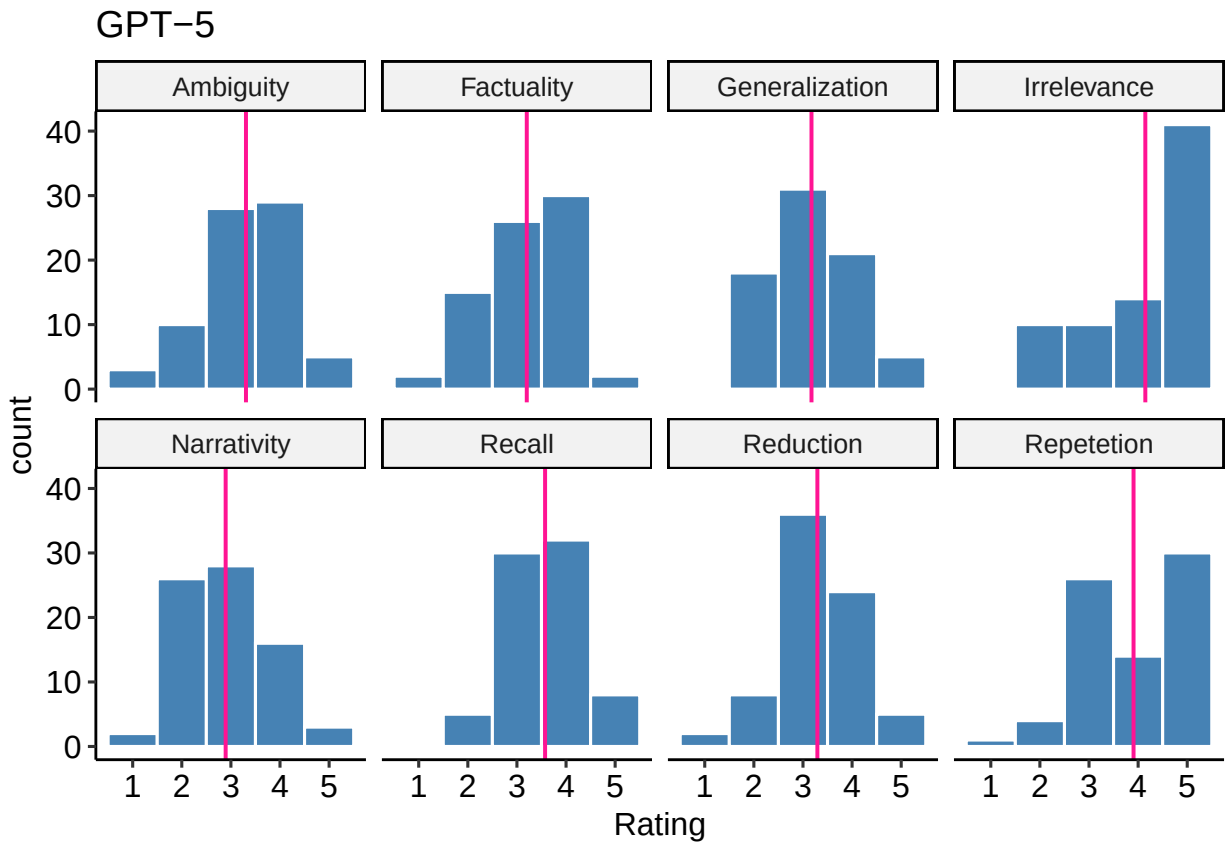
plot(filter(complete, Model=="GPT4"), "GPT-4")
```

```
## Warning: Removed 273 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
plot(filter(complete, Model=="GPT5"), "GPT-5")
```

```
## Warning: Removed 272 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
# Highest rated questions
complete.filtered %>%
  group_by(Question, Model) %>%
  summarise(M_Rating = mean(Rating, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = Model, values_from = M_Rating) %>%
  mutate(M_Overall = (GPT4 + GPT5) / 2) %>%
  arrange(desc(M_Overall)) %>%
  print(n = 24)
```

```
## # A tibble: 24 x 4
##    Question  GPT4  GPT5 M_Overall
##    <chr>    <dbl> <dbl>     <dbl>
##  1 Q9        3.81  4         3.91
##  2 Q10       3.53  3.88      3.70
##  3 Q3        3.53  3.79      3.66
##  4 Q2        3.48  3.78      3.63
##  5 Q5        3.19  3.96      3.57
##  6 Q21       3.47  3.67      3.57
##  7 Q12       3.66  3.46      3.56
##  8 Q8        3.88  3.21      3.54
##  9 Q19       3.41  3.58      3.49
## 10 Q6        3.5   3.46      3.48
## 11 Q17       3.38  3.54      3.46
## 12 Q16       3.47  3.42      3.44
## 13 Q15       3.12  3.75      3.44
## 14 Q11       3.53  3.33      3.43
```

```
## 15 Q1        3.25  3.59      3.42
## 16 Q24       3.62  3.17      3.40
## 17 Q18       3.44  3.29      3.36
## 18 Q14       3.44  3.17      3.30
## 19 Q4        3.25  3.34      3.30
## 20 Q13       3.66  2.83      3.24
## 21 Q7        3.19  3.21      3.20
## 22 Q23       3.22  2.88      3.05
## 23 Q22       2.88  3.08      2.98
## 24 Q20       1.75  2.96      2.35
```

```r
# Rater means and sd per model per criteria
complete.filtered %>%
  filter(!is.na(Rating)) %>%
  group_by(Rater, Model, Criteria) %>%
  summarise(mean_rating = mean(Rating), sd_rating = sd(Rating)) %>%
  print(n = 80)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Rater, Model, and Criteria.
## i Output is grouped by Rater and Model.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Rater, Model, Criteria))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 88 x 5
## # Groups:   Rater, Model [11]
##    Rater Model Criteria       mean_rating sd_rating
##    <chr> <chr> <chr>                <dbl>     <dbl>
##  1 A     GPT4  Ambiguity             3.67     0.816
##  2 A     GPT4  Factuality            3.2      0.775
##  3 A     GPT4  Generalization        3.07     0.799
##  4 A     GPT4  Irrelevance           4.47     0.990
##  5 A     GPT4  Narrativity           3.93     0.799
##  6 A     GPT4  Recall                3.2      0.862
##  7 A     GPT4  Reduction             3.27     0.594
##  8 A     GPT4  Repetetion            4        1.07
##  9 A     GPT5  Ambiguity             3.53     0.915
## 10 A     GPT5  Factuality            3.27     0.704
## 11 A     GPT5  Generalization        2.93     0.704
## 12 A     GPT5  Irrelevance           4.27     0.961
## 13 A     GPT5  Narrativity           3.47     0.743
## 14 A     GPT5  Recall                3.2      0.676
## 15 A     GPT5  Reduction             3.27     0.594
## 16 A     GPT5  Repetetion            4.13     1.06
## 17 Ana   GPT4  Ambiguity             3.17     0.963
## 18 Ana   GPT4  Factuality            2.79     1.02
## 19 Ana   GPT4  Generalization        2.83     0.761
## 20 Ana   GPT4  Irrelevance           4.29     1.16
## 21 Ana   GPT4  Narrativity           3.17     0.868
## 22 Ana   GPT4  Recall                3.42     0.974
## 23 Ana   GPT4  Reduction             2.79     0.721
## 24 Ana   GPT4  Repetetion            3.5      1.38
```

```
## 25 K     GPT4  Ambiguity            3.07     0.829
## 26 K     GPT4  Factuality           2.93     0.730
## 27 K     GPT4  Generalization       2.93     0.730
## 28 K     GPT4  Irrelevance          4.36     0.842
## 29 K     GPT4  Narrativity          3.71     0.611
## 30 K     GPT4  Recall               3.43     0.646
## 31 K     GPT4  Reduction            3.07     0.616
## 32 K     GPT4  Repetetion           3.85     1.07
## 33 K     GPT5  Ambiguity            3.29     0.611
## 34 K     GPT5  Factuality           2.86     0.864
## 35 K     GPT5  Generalization       2.93     0.829
## 36 K     GPT5  Irrelevance          3.86     1.03
## 37 K     GPT5  Narrativity          3.5      0.519
## 38 K     GPT5  Recall               3.43     0.646
## 39 K     GPT5  Reduction            3.21     0.699
## 40 K     GPT5  Repetetion           3.71     0.994
## 41 L     GPT4  Ambiguity            2.62     0.719
## 42 L     GPT4  Factuality           2        0.966
## 43 L     GPT4  Generalization       3.75     1
## 44 L     GPT4  Irrelevance          3.69     1.40
## 45 L     GPT4  Narrativity          3.06     0.929
## 46 L     GPT4  Recall               2.38     1.02
## 47 L     GPT4  Reduction            2.81     0.981
## 48 L     GPT4  Repetetion           4        1.03
## 49 L     GPT5  Ambiguity            2.44     1.09
## 50 L     GPT5  Factuality           2.44     0.814
## 51 L     GPT5  Generalization       3.06     1.12
## 52 L     GPT5  Irrelevance          3.06     1.29
## 53 L     GPT5  Narrativity          2        0.516
## 54 L     GPT5  Recall               3.06     0.680
## 55 L     GPT5  Reduction            2.69     0.946
## 56 L     GPT5  Repetetion           3.69     1.25
## 57 S     GPT4  Ambiguity            3.08     0.494
## 58 S     GPT4  Factuality           3.31     0.630
## 59 S     GPT4  Generalization       2.92     0.641
## 60 S     GPT4  Irrelevance          4.69     0.480
## 61 S     GPT4  Narrativity          3.92     0.760
## 62 S     GPT4  Recall               4.31     0.480
## 63 S     GPT4  Reduction            3.31     0.751
## 64 S     GPT4  Repetetion           3.77     1.01
## 65 S     GPT5  Ambiguity            3.85     0.555
## 66 S     GPT5  Factuality           3.62     0.506
## 67 S     GPT5  Generalization       3.62     0.506
## 68 S     GPT5  Irrelevance          4.85     0.376
## 69 S     GPT5  Narrativity          2.77     0.599
## 70 S     GPT5  Recall               4.15     0.689
## 71 S     GPT5  Reduction            4        0.707
## 72 S     GPT5  Repetetion           4.23     0.927
## 73 Y     GPT4  Ambiguity            3.24     0.664
## 74 Y     GPT4  Factuality           2.94     0.827
## 75 Y     GPT4  Generalization       2.88     0.697
## 76 Y     GPT4  Irrelevance          4.41     1.06
## 77 Y     GPT4  Narrativity          3.24     0.562
## 78 Y     GPT4  Recall               3.47     0.717
```

```
## 79 Y     GPT4  Reduction             2.88     0.332
## 80 Y     GPT4  Repetetion            3.59     1.18
## # i 8 more rows
```

Inference

Krippendorf's Alpha

- test for inter-rater agreement

```
check_without_rater <- function(rater) {
  filtered <- complete.filtered %>% filter(Rater != rater)

  cat("Original rows:", nrow(complete.filtered), "\n")
  cat("Filtered rows:", nrow(filtered), "\n")
  cat("Remaining raters:", paste(unique(filtered$Rater), collapse = ", "), "\n\n")

  kripp.results <- filtered %>%
  group_by(Model, Criteria) %>%
  group_modify(~{
    mat <- .x %>%
      pivot_wider(names_from = Rater, values_from = Rating) %>%
      select(-Question) %>%
      as.matrix()

    result <- kripp.alpha(t(mat), method = "ordinal")
    tibble(alpha = result$value)
  }) %>%
    mutate(excluded = rater)

  kripp.results.excl.overall <- filtered %>%
  group_by(Model) %>%
  group_modify(~ {
    mat <- .x %>%
      pivot_wider(names_from = c(Rater), values_from = Rating,
                  id_cols = c(Question, Criteria)) %>%
      select(-c(Question, Criteria)) %>%
      as.matrix()
    result <- kripp.alpha(t(mat), method = "ordinal")
    tibble(alpha = result$value)
  })

  cat("=== Excluding", rater, "===\n\n")
  cat("--- Per Model and Criteria ---\n")
  print(kripp.results)
  cat("\n--- Overall per Model ---\n")
  print(kripp.results.excl.overall)

  return(kripp.results)
}

## exclude manually
# check_without_rater("A")
```

```r
# check_without_rater("K")
# check_without_rater("L")
# check_without_rater("S")
# check_without_rater("Y")
# check_without_rater("")


# 1. Baseline (outside function, already computed earlier)
kripp.results.overall <- complete.filtered %>%
  group_by(Model) %>%
  group_modify(~ {
    mat <- .x %>%
      pivot_wider(names_from = c(Rater), values_from = Rating,
                  id_cols = c(Question, Criteria)) %>%
      select(-c(Question, Criteria)) %>%
      as.matrix()

    result <- kripp.alpha(t(mat), method = "ordinal")
    tibble(alpha = result$value)
  })

cat("=== Baseline Overall Alpha ===\n")
```

```
## === Baseline Overall Alpha ===
```

```r
print(kripp.results.overall)
```

```
## # A tibble: 2 x 2
## # Groups:   Model [2]
##   Model alpha
##   <chr> <dbl>
## 1 GPT4  0.372
## 2 GPT5  0.117
```

```r
# 2. Per criteria alpha
kripp.results <- complete.filtered %>%
  group_by(Model, Criteria) %>%
  group_modify(~{
    mat <- .x %>%
      pivot_wider(names_from = Rater, values_from = Rating) %>%
      select(-Question) %>%
      as.matrix()

    result <- kripp.alpha(t(mat), method = "ordinal")
    tibble(alpha = result$value)
  })

cat("=== Per Model and Criteria ===\n")
```

```
## === Per Model and Criteria ===
```

```r
print(kripp.results)
```

```
## # A tibble: 16 x 3
## # Groups:   Model, Criteria [16]
##    Model Criteria         alpha
##    <chr> <chr>            <dbl>
##  1 GPT4  Ambiguity      -0.0335
##  2 GPT4  Factuality      0.114
##  3 GPT4  Generalization  0.217
##  4 GPT4  Irrelevance     0.302
##  5 GPT4  Narrativity     0.0719
##  6 GPT4  Recall          0.298
##  7 GPT4  Reduction       0.134
##  8 GPT4  Repetetion      0.543
##  9 GPT5  Ambiguity      -0.112
## 10 GPT5  Factuality     -0.0580
## 11 GPT5  Generalization  0.0148
## 12 GPT5  Irrelevance     0.0725
## 13 GPT5  Narrativity    -0.186
## 14 GPT5  Recall         -0.0846
## 15 GPT5  Reduction       0.0254
## 16 GPT5  Repetetion      0.176
```

```r
# 2. Sensitivity analysis
exclusion_resutls <- map_dfr(c("A", "K", "L", "S", "Y", "Ana"), check_without_rater)
```

```
## Original rows: 1391
## Filtered rows: 1151
## Remaining raters: Ana, K, L, S, Y
##
## === Excluding A ===
##
## --- Per Model and Criteria ---
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria          alpha excluded
##    <chr> <chr>             <dbl> <chr>
##  1 GPT4  Ambiguity       0.0591  A
##  2 GPT4  Factuality      0.0705  A
##  3 GPT4  Generalization  0.249   A
##  4 GPT4  Irrelevance     0.277   A
##  5 GPT4  Narrativity    -0.0252  A
##  6 GPT4  Recall          0.203   A
##  7 GPT4  Reduction       0.229   A
##  8 GPT4  Repetetion      0.476   A
##  9 GPT5  Ambiguity      -0.127   A
## 10 GPT5  Factuality      0.00402 A
## 11 GPT5  Generalization  0.0905  A
## 12 GPT5  Irrelevance     0.0764  A
## 13 GPT5  Narrativity    -0.329   A
## 14 GPT5  Recall         -0.0512  A
## 15 GPT5  Reduction      -0.0708  A
```

```
## 16 GPT5  Repetetion      0.130   A
##
## --- Overall per Model ---
## # A tibble: 2 x 2
## # Groups:   Model [2]
##   Model alpha
##   <chr> <dbl>
## 1 GPT4  0.356
## 2 GPT5  0.123
## Original rows: 1391
## Filtered rows: 1168
## Remaining raters: A, Ana, L, S, Y
##
## === Excluding K ===
##
## --- Per Model and Criteria ---
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria          alpha excluded
##    <chr> <chr>             <dbl> <chr>
##  1 GPT4  Ambiguity       0.00551 K
##  2 GPT4  Factuality      0.118   K
##  3 GPT4  Generalization  0.317   K
##  4 GPT4  Irrelevance     0.282   K
##  5 GPT4  Narrativity     0.134   K
##  6 GPT4  Recall          0.368   K
##  7 GPT4  Reduction       0.137   K
##  8 GPT4  Repetetion      0.608   K
##  9 GPT5  Ambiguity      -0.187   K
## 10 GPT5  Factuality     -0.0820  K
## 11 GPT5  Generalization  0.00905 K
## 12 GPT5  Irrelevance     0.153   K
## 13 GPT5  Narrativity    -0.272   K
## 14 GPT5  Recall         -0.0206  K
## 15 GPT5  Reduction       0.109   K
## 16 GPT5  Repetetion      0.181   K
##
## --- Overall per Model ---
## # A tibble: 2 x 2
## # Groups:   Model [2]
##   Model alpha
##   <chr> <dbl>
## 1 GPT4  0.403
## 2 GPT5  0.124
## Original rows: 1391
## Filtered rows: 1135
## Remaining raters: A, Ana, K, S, Y
##
## === Excluding L ===
##
## --- Per Model and Criteria ---
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria         alpha excluded
```

```
##    <chr> <chr>            <dbl> <chr>
##  1 GPT4  Ambiguity       0.0104 L
##  2 GPT4  Factuality      0.449  L
##  3 GPT4  Generalization  0.143  L
##  4 GPT4  Irrelevance     0.601  L
##  5 GPT4  Narrativity     0.0860 L
##  6 GPT4  Recall          0.606  L
##  7 GPT4  Reduction       0.187  L
##  8 GPT4  Repetetion      0.725  L
##  9 GPT5  Ambiguity      -0.0354 L
## 10 GPT5  Factuality     -0.127  L
## 11 GPT5  Generalization -0.121  L
## 12 GPT5  Irrelevance     0.236  L
## 13 GPT5  Narrativity    -0.0446 L
## 14 GPT5  Recall         -0.0537 L
## 15 GPT5  Reduction       0.175  L
## 16 GPT5  Repetetion      0.365  L
##
## --- Overall per Model ---
## # A tibble: 2 x 2
## # Groups:   Model [2]
##   Model alpha
##   <chr> <dbl>
## 1 GPT4  0.535
## 2 GPT5  0.216
## Original rows: 1391
## Filtered rows: 1183
## Remaining raters: A, Ana, K, L, Y
##
## === Excluding S ===
##
## --- Per Model and Criteria ---
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria         alpha excluded
##    <chr> <chr>            <dbl> <chr>
##  1 GPT4  Ambiguity      -0.0151 S
##  2 GPT4  Factuality      0.0459 S
##  3 GPT4  Generalization  0.190  S
##  4 GPT4  Irrelevance     0.265  S
##  5 GPT4  Narrativity     0.0960 S
##  6 GPT4  Recall          0.206  S
##  7 GPT4  Reduction       0.145  S
##  8 GPT4  Repetetion      0.501  S
##  9 GPT5  Ambiguity      -0.0280 S
## 10 GPT5  Factuality     -0.225  S
## 11 GPT5  Generalization  0.0674 S
## 12 GPT5  Irrelevance    -0.0182 S
## 13 GPT5  Narrativity    -0.205  S
## 14 GPT5  Recall         -0.192  S
## 15 GPT5  Reduction       0.0391 S
## 16 GPT5  Repetetion      0.209  S
##
## --- Overall per Model ---
```

```
## # A tibble: 2 x 2
## # Groups:   Model [2]
##   Model  alpha
##   <chr>  <dbl>
## 1 GPT4  0.336
## 2 GPT5  0.0740
## Original rows: 1391
## Filtered rows: 1119
## Remaining raters: A, Ana, K, L, S
##
## === Excluding Y ===
##
## --- Per Model and Criteria ---
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria          alpha excluded
##    <chr> <chr>             <dbl> <chr>
##  1 GPT4  Ambiguity      -0.0914  Y
##  2 GPT4  Factuality      0.0347  Y
##  3 GPT4  Generalization  0.175   Y
##  4 GPT4  Irrelevance     0.240   Y
##  5 GPT4  Narrativity     0.0394  Y
##  6 GPT4  Recall          0.217   Y
##  7 GPT4  Reduction       0.0702  Y
##  8 GPT4  Repetetion      0.480   Y
##  9 GPT5  Ambiguity      -0.251   Y
## 10 GPT5  Factuality      0.0322  Y
## 11 GPT5  Generalization -0.0184  Y
## 12 GPT5  Irrelevance    -0.0725  Y
## 13 GPT5  Narrativity    -0.199   Y
## 14 GPT5  Recall         -0.203   Y
## 15 GPT5  Reduction      -0.154   Y
## 16 GPT5  Repetetion      0.00517 Y
##
## --- Overall per Model ---
## # A tibble: 2 x 2
## # Groups:   Model [2]
##   Model  alpha
##   <chr>  <dbl>
## 1 GPT4  0.315
## 2 GPT5  0.0229
## Original rows: 1391
## Filtered rows: 1199
## Remaining raters: A, K, L, S, Y
##
## === Excluding Ana ===
##
## --- Per Model and Criteria ---
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria         alpha excluded
##    <chr> <chr>            <dbl> <chr>
##  1 GPT4  Ambiguity      -0.209  Ana
##  2 GPT4  Factuality     -0.0299 Ana
```

```
##  3 GPT4  Generalization  0.205  Ana
##  4 GPT4  Irrelevance     0.176  Ana
##  5 GPT4  Narrativity     0.0617 Ana
##  6 GPT4  Recall          0.194  Ana
##  7 GPT4  Reduction       0.0382 Ana
##  8 GPT4  Repetetion      0.448  Ana
##  9 GPT5  Ambiguity      -0.112  Ana
## 10 GPT5  Factuality     -0.0580 Ana
## 11 GPT5  Generalization  0.0148 Ana
## 12 GPT5  Irrelevance     0.0725 Ana
## 13 GPT5  Narrativity    -0.186  Ana
## 14 GPT5  Recall         -0.0846 Ana
## 15 GPT5  Reduction       0.0254 Ana
## 16 GPT5  Repetetion      0.176  Ana
##
## --- Overall per Model ---
## # A tibble: 2 x 2
## # Groups:   Model [2]
##   Model alpha
##   <chr> <dbl>
## 1 GPT4  0.288
## 2 GPT5  0.117
```

```
exclusion_resutls %>%
  group_by(Model, Criteria) %>%
  slice_max(alpha, n = 1) %>%
  select(Model, Criteria, best_exclusion = excluded, alpha)
```

```
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria       best_exclusion   alpha
##    <chr> <chr>          <chr>            <dbl>
##  1 GPT4  Ambiguity      A               0.0591
##  2 GPT4  Factuality     L               0.449
##  3 GPT4  Generalization K               0.317
##  4 GPT4  Irrelevance    L               0.601
##  5 GPT4  Narrativity    K               0.134
##  6 GPT4  Recall         L               0.606
##  7 GPT4  Reduction      A               0.229
##  8 GPT4  Repetetion     L               0.725
##  9 GPT5  Ambiguity      S              -0.0280
## 10 GPT5  Factuality     Y               0.0322
## 11 GPT5  Generalization A               0.0905
## 12 GPT5  Irrelevance    L               0.236
## 13 GPT5  Narrativity    L              -0.0446
## 14 GPT5  Recall         K              -0.0206
## 15 GPT5  Reduction      L               0.175
## 16 GPT5  Repetetion     L               0.365
```

Weighted Kappa

- weighted kappa (quadratic) as robustness check for Krippendorff's alpha
- it can only look at 2 rater pairs at once

- computed pairwise across all rater combinations (15 pairs from 6 raters), then averaged
- quadratic weights match those used in ordinal alpha

```
raters <- c("A", "Ana", "K", "L", "S", "Y")
pairs <- combn(raters, 2, simplify = FALSE)

weighted.kappa.results <- complete.filtered %>%
  group_by(Model, Criteria) %>%
  group_modify(~ {
    kappas <- map_dbl(pairs, function(pair) {
      d <- .x %>%
        filter(Rater %in% pair) %>%
        pivot_wider(names_from = Rater, values_from = Rating, id_cols = Question)

      # skip if either rater has no data in this group
      if (!all(pair %in% names(d))) return(NA)

      d <- d %>% select(all_of(pair)) %>% na.omit()

      if (nrow(d) < 2) return(NA)
      kappa2(d, weight = "squared")$value
    })
    tibble(kappa = mean(kappas, na.rm = TRUE))
  })

print(weighted.kappa.results)
```

```
## # A tibble: 16 x 3
## # Groups:   Model, Criteria [16]
##    Model Criteria         kappa
##    <chr> <chr>            <dbl>
##  1 GPT4  Ambiguity      -0.0544
##  2 GPT4  Factuality       0.148
##  3 GPT4  Generalization   0.127
##  4 GPT4  Irrelevance      0.358
##  5 GPT4  Narrativity    -0.0147
##  6 GPT4  Recall           0.240
##  7 GPT4  Reduction        0.0733
##  8 GPT4  Repetetion       0.448
##  9 GPT5  Ambiguity        0.0313
## 10 GPT5  Factuality     -0.0930
## 11 GPT5  Generalization -0.112
## 12 GPT5  Irrelevance      0.0942
## 13 GPT5  Narrativity    -0.112
## 14 GPT5  Recall         -0.0910
## 15 GPT5  Reduction        0.0248
## 16 GPT5  Repetetion       0.142
```

```
comparison <- kripp.results %>%
  left_join(weighted.kappa.results, by = c("Model", "Criteria")) %>%
```

```
  arrange(Model, Criteria)

print(comparison)
```

Agreement comparision – alpha, kappa

```
## # A tibble: 16 x 4
## # Groups:   Model, Criteria [16]
##    Model Criteria         alpha   kappa
##    <chr> <chr>            <dbl>   <dbl>
##  1 GPT4  Ambiguity      -0.0335 -0.0544
##  2 GPT4  Factuality       0.114   0.148
##  3 GPT4  Generalization   0.217   0.127
##  4 GPT4  Irrelevance      0.302   0.358
##  5 GPT4  Narrativity      0.0719 -0.0147
##  6 GPT4  Recall           0.298   0.240
##  7 GPT4  Reduction        0.134   0.0733
##  8 GPT4  Repetetion       0.543   0.448
##  9 GPT5  Ambiguity      -0.112   0.0313
## 10 GPT5  Factuality     -0.0580 -0.0930
## 11 GPT5  Generalization   0.0148 -0.112
## 12 GPT5  Irrelevance      0.0725  0.0942
## 13 GPT5  Narrativity     -0.186  -0.112
## 14 GPT5  Recall          -0.0846 -0.0910
## 15 GPT5  Reduction        0.0254  0.0248
## 16 GPT5  Repetetion       0.176   0.142
```

Output correlation data

```
make_criteria_cor <- function(data) {
  criteria_means <- data %>%
    filter(Model %in% c("GPT4", "GPT5"), !is.na(Rating)) %>%
    group_by(Question, Criteria) %>%
    summarise(mean_rating = mean(Rating, na.rm = TRUE), .groups = "drop") %>%
    pivot_wider(names_from = Criteria, values_from = mean_rating) %>%
    select(-Question)

  cor_mat <- cor(criteria_means, use = "pairwise.complete.obs")
  cor_mat[upper.tri(cor_mat)] <- NA

  labels <- colnames(cor_mat)

  # Convert to plain list of lists - no names anywhere
  z <- lapply(seq_len(nrow(cor_mat)), function(i) {
  row <- as.numeric(cor_mat[i, ])
  lapply(row, function(x) if (is.na(x)) NA_real_ else x)
})

  list(labels = labels, z = z)
}
```

```r
output <- list(
  excluding_auth = make_criteria_cor(complete %>% filter(Rater != "Ana")),
  including_auth = make_criteria_cor(complete)
)

write_json(output, "plots/criteria_correlation.json", digits = 4, pretty = TRUE, auto_unbox = TRUE)
```

```r
make_criteria_cor_plot <- function(data, title = NULL) {
  criteria_means <- data %>%
    filter(Model %in% c("GPT4", "GPT5"), !is.na(Rating)) %>%
    group_by(Question, Criteria) %>%
    summarise(mean_rating = mean(Rating, na.rm = TRUE), .groups = "drop") %>%
    pivot_wider(names_from = Criteria, values_from = mean_rating) %>%
    select(-Question)

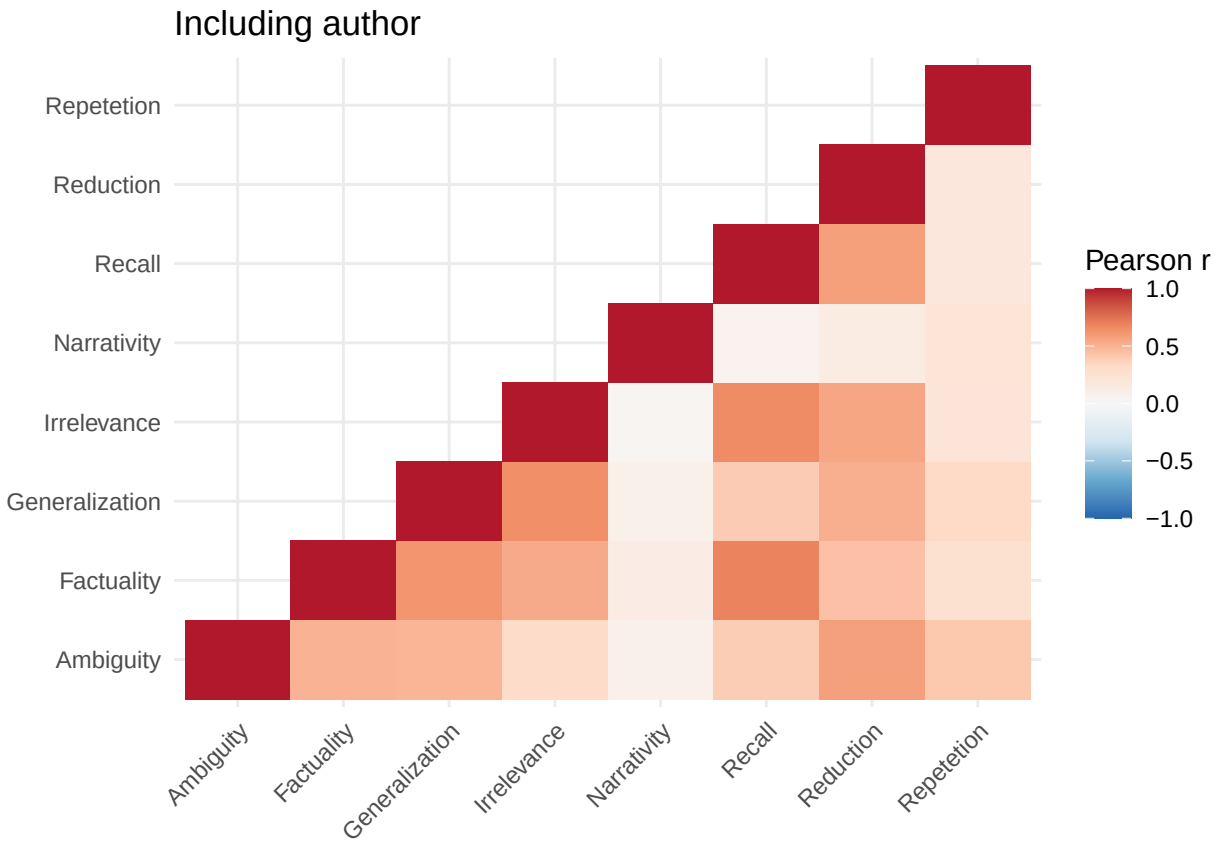
  cor_mat <- cor(criteria_means, use = "pairwise.complete.obs")
  cor_mat[upper.tri(cor_mat)] <- NA

  melted <- as.data.frame(cor_mat) %>%
    rownames_to_column("Var1") %>%
    pivot_longer(-Var1, names_to = "Var2", values_to = "value") %>%
    filter(!is.na(value))

  ggplot(melted, aes(Var1, Var2, fill = value)) +
    geom_tile() +
    scale_fill_distiller(palette = "RdBu", limits = c(-1, 1), name = "Pearson r") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
    labs(x = NULL, y = NULL, title = title)
}

p1 <- make_criteria_cor_plot(complete %>% filter(Rater != "Ana"), "Excluding author")
p2 <- make_criteria_cor_plot(complete, "Including author")

p2
```




```r
make_rater_scatter <- function(data, title = NULL) {
  d <- data %>%
    filter(!is.na(Rating), Model != "RAG") %>%
    group_by(Model, Rater, Criteria) %>%
    summarise(mean_rating = mean(Rating, na.rm = TRUE), .groups = "drop")

  raters <- unique(d$Rater)
  rater_pairs <- combn(raters, 2, simplify = FALSE)

  pair_data <- lapply(rater_pairs, function(pair) {
    d %>%
      filter(Rater %in% pair) %>%
      pivot_wider(names_from = Rater, values_from = mean_rating) %>%
      rename(Rater1_val = all_of(pair[1]), Rater2_val = all_of(pair[2])) %>%
      mutate(Rater1 = pair[1], Rater2 = pair[2])
  }) %>% bind_rows()

  ggplot(pair_data, aes(x = Rater1_val, y = Rater2_val, color = Criteria, shape = Model)) +
    geom_point(size = 5, alpha = 0.8) +
    geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey50") +
    facet_wrap(~paste(Rater1, "-", Rater2)) +
    scale_x_continuous(limits = c(1, 5)) +
    scale_y_continuous(limits = c(1, 5)) +
    theme_minimal(base_size = 16) +
    theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
    labs(x = "Mean rating (rater 1)", y = "Mean rating (rater 2)", title = title)
```

```
}

p <- make_rater_scatter(complete)
ggsave("plots/rater_scatter.pdf", p, width = 16, height = 14)
```

```
## Warning: Removed 40 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'L - S' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'L - Y' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'S - Y' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'Ana - Y' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'K - L' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'K - S' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'K - Y' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'A - Y' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'Ana - K' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'Ana - L' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'Ana - S' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'A - Ana' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'A - K' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'A - L' in 'mbcsToSbcs': - substituted for – (U+2013)

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## for 'A - S' in 'mbcsToSbcs': - substituted for – (U+2013)
```

Paired t-test

- test for model differences

```
# make model ratings into wide format (for t-test)
complete_wide <- complete %>%
  group_by(Rater, Question, Criteria, Model) %>%
  summarise(Rating = mean(Rating, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(
    names_from = Model,
    values_from = Rating
  )

head(complete_wide)
```

```
## # A tibble: 6 x 6
##   Rater Question Criteria        GPT4  GPT5   RAG
##   <chr> <chr>    <chr>          <dbl> <dbl> <dbl>
## 1 A     Q1       Ambiguity          4     4     5
## 2 A     Q1       Factuality         4     3     5
## 3 A     Q1       Generalization     3     2     5
## 4 A     Q1       Irrelevance        5     5     5
## 5 A     Q1       Narrativity        3     3     5
## 6 A     Q1       Recall             2     4     5
```

```
# Summarise by model
complete_wide %>%
  summarise(
    M_GPT4 = mean(GPT4, na.rm=TRUE), SD_GPT4 = sd(GPT4, na.rm=TRUE),
    M_GPT5 = mean(GPT5, na.rm=TRUE), SD_GPT5 = sd(GPT5, na.rm=TRUE)
  )
```

```
## # A tibble: 1 x 4
##   M_GPT4 SD_GPT4 M_GPT5 SD_GPT5
##    <dbl>   <dbl>  <dbl>   <dbl>
## 1   3.36    1.02   3.44   0.999
```

```
# overall
t.test(complete_wide$GPT4, complete_wide$GPT5, paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  complete_wide$GPT4 and complete_wide$GPT5
## t = -0.95701, df = 598, p-value = 0.339
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -0.13248150  0.04567015
## sample estimates:
## mean difference
##     -0.04340568
```

```r
# group differences across raters
t <- complete_wide %>%
  filter(!is.na(GPT4), !is.na(GPT5)) %>%
  group_by(Rater) %>%
  summarise(
    t_test = list(t.test(GPT4, GPT5, paired = TRUE)),
    .groups = "drop"
  )

# cohen's d
d_values <- complete_wide %>%
  group_by(Criteria) %>%
  summarise(
    cohens_d = mean(GPT4 - GPT5, na.rm = TRUE) / sd(GPT4 - GPT5, na.rm = TRUE),
    .groups = "drop"
  )

# per criterion
t_criteria <- complete_wide %>%
  filter(Criteria != "Irrelevance") %>% # correction will apply for 7 tests
  group_by(Criteria) %>%
  summarise(
    M_GPT4    = mean(GPT4, na.rm = TRUE),
    SD_GPT4   = sd(GPT4, na.rm = TRUE),
    M_GPT5    = mean(GPT5, na.rm = TRUE),
    SD_GPT5   = sd(GPT5, na.rm = TRUE),
    test      = list(t.test(GPT4, GPT5, paired = TRUE)),
    .groups   = "drop"
  ) %>%
  mutate(
    t_stat    = map_dbl(test, ~ .x$statistic),
    df        = map_dbl(test, ~ .x$parameter),
    p_value   = map_dbl(test, ~ .x$p.value),
    p_bonferroni = p.adjust(p_value, method = "bonferroni"),
    mean_diff = map_dbl(test, ~ .x$estimate),
    ci_low    = map_dbl(test, ~ .x$conf.int[1]),
    ci_high   = map_dbl(test, ~ .x$conf.int[2])
  ) %>%
  left_join(d_values, by = "Criteria") %>%
  select(-test)

print(t_criteria)
```

```
## # A tibble: 7 x 13
##   Criteria       M_GPT4 SD_GPT4 M_GPT5 SD_GPT5 t_stat    df p_value p_bonferroni
##   <chr>           <dbl>   <dbl>  <dbl>   <dbl>  <dbl> <dbl>   <dbl>        <dbl>
## 1 Ambiguity        3.14   0.821   3.31   0.930 -1.54     74 1.29e-1    0.903
## 2 Factuality       2.84   0.934   3.2    0.885 -3.20     74 2.00e-3    0.0140
## 3 Generalization   3.05   0.825   3.17   0.876 -0.379    74 7.06e-1    1
## 4 Narrativity      3.45   0.836   2.89   0.909  5.22     74 1.59e-6    0.0000111
## 5 Recall           3.34   0.971   3.57   0.774 -2.13     74 3.61e-2    0.253
## 6 Reduction        2.99   0.707   3.29   0.851 -2.06     74 4.34e-2    0.304
## 7 Repetetion       3.76   1.15    3.91   1.04  -0.740    73 4.62e-1    1
```

```
## # i 4 more variables: mean_diff <dbl>, ci_low <dbl>, ci_high <dbl>,
## #   cohens_d <dbl>
```

```
print(d_values)
```

```
## # A tibble: 8 x 2
##   Criteria       cohens_d
##   <chr>             <dbl>
## 1 Ambiguity       -0.177
## 2 Factuality      -0.370
## 3 Generalization  -0.0438
## 4 Irrelevance      0.132
## 5 Narrativity      0.603
## 6 Recall          -0.246
## 7 Reduction       -0.237
## 8 Repetetion      -0.0860
```

Principal Component Analysis

- test to see if criteria dimensions can be reduced

```
# if there is time
```